

## Chapter 23

### Brown-out Detector

#### 23.1 Introduction

The brown-out detector of ESP32-P4 monitors the voltage levels of pins VDD\_ANA and VDD\_BAT. If the voltage on these pins drops below the predefined threshold (defaulting to 2.4 V), the detector triggers signals to shut down certain power-consuming blocks (e.g., flash), ensuring that the digital module has sufficient time to save and transfer important data.

#### 23.2 Feature List

- Two monitored sources
  - Monitors the voltage level of pin VDD\_ANA, which supplies power to the analog circuit.
  - Monitors the voltage level of pin VDD\_BAT, which connects the external battery to the chip.
- Two configurable monitoring modes
  - Mode 0: The brown-out detector triggers interrupts when the brown-out counter reaches the predefined threshold and selects the reset mode according to the configuration.
  - Mode 1: The brown-out detector triggers a system reset when the voltage falls below the threshold.
- Configurable voltage-monitoring thresholds and noise tolerance
- Configurable handling modes for under-voltage events

## 23.3 Architectural Overview

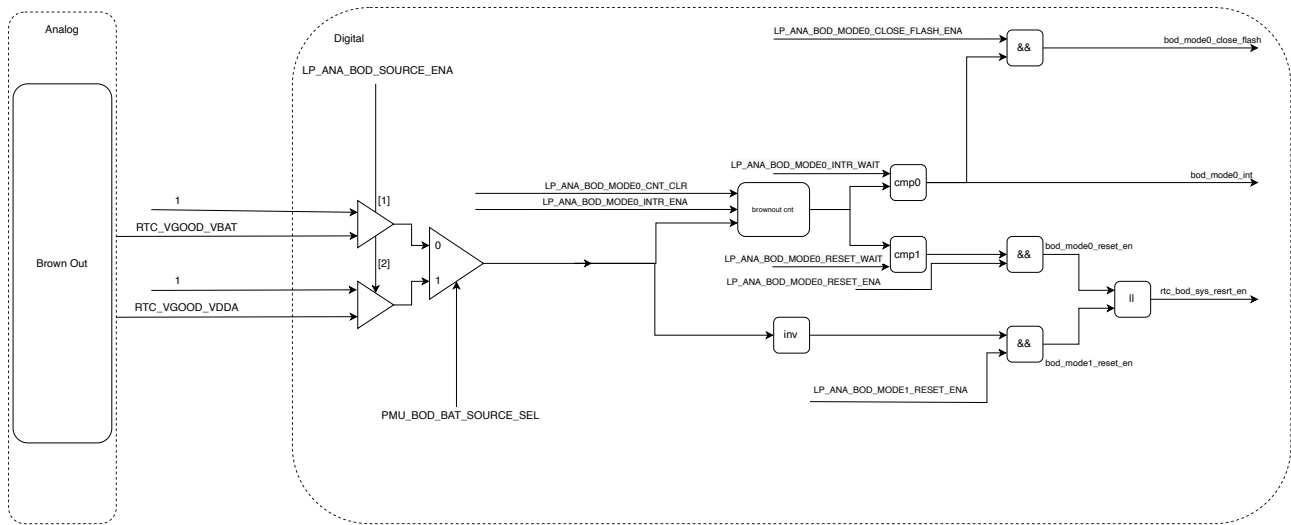


Figure 23.3-1. Brown-out Detector Overview

- Source selection for detection: The brown-out detector can select which power source to monitor by configuring `LP_ANA_BOD_SOURCE_ENA`. `RTC_VGOOD_VBAT` indicates the detection result of `VDD_BAT`, and `RTC_VGOOD_VDDA` indicates the detection result of `VDD_ANA`.
- Brown-out detector (Brown Out): An analog circuit designed to monitor the voltage of power pins and generate brown-out signals. It compares the voltage with the predefined threshold. When the voltage falls below this threshold, the detector outputs brown-out signals.
- Brown-out counter (brownout cnt): A counter used for filtering voltage noise on the detected power pins. When the brown-out signal is received, the counter starts counting based on the clock source `LP_DYN_FAST_CLK`. If the brown-out signal ceases during the counting process, the counter resets to 0. This counter can be cleared by configuring `LP_ANA_BOD_MODE0_CNT_CLR`.
- Interrupt comparator (cmp0): The comparator generates interrupts by comparing the counter value with the threshold. When the counter value exceeds the threshold, the comparator generates the interrupt signal `bod_mode0_int`. If `LP_ANA_BOD_MODE0_CLOSE_FLASH_ENA` is enabled, `bod_mode0_close_flash` will be triggered, causing flash to enter the SUSPEND state.
- Reset comparator (cmp1): The comparator triggers reset signals by comparing the counter value with the threshold. When the counter value exceeds the threshold, the comparator triggers reset signals.

## 23.4 Functional Description

- Monitored sources
  - `VDD_ANA`: Power pin for the analog circuit, supporting both Mode 0 and Mode 1.
  - `VDD_BAT`: Power pin connected to the battery, supporting both Mode 0 and Mode 1. In addition, it supports two configurable voltage thresholds:
    - \* `VBAT_BOD`: Used for brown-out detection of `VDD_BAT`

\* VBAT\_CHARGER: Indicates whether the battery connected to VDD\_BAT needs charging.

These thresholds trigger different interrupts.

- Monitoring modes

There are two monitoring modes based on the way the brown-out detector handles the brown-out signal:

- Mode 0: When Mode 0 is enabled, the brown-out counter starts counting after detecting a brown-out signal. Interrupt and reset signals are generated when the counter reaches the thresholds defined by [LP\\_ANA\\_BOD\\_MODE0\\_RESET\\_WAIT](#) and [LP\\_ANA\\_BOD\\_MODE0\\_INTR\\_WAIT](#), respectively. This helps filter out noise on the monitored pins. Configure [LP\\_ANA\\_BOD\\_MODE0\\_RESET\\_SEL](#) to select the reset options:
  - \* 0: Reset the chip.
  - \* 1: Reset the system.
- Mode 1: Mode 1 is the default mode, which resets the system immediately upon detecting under-voltage.

- Voltage-monitoring thresholds

The voltage-monitoring threshold is configured in the analog circuit by the analog I2C master. The threshold configuration registers are as follows. BIAS is the slave of the analog I2C master. The address of BIAS is 0x6a. REGXX indicates the register address of BIAS, and [x:x] indicates the bits of this register.

- BIAS\_OR\_DREFL\_VBAT (REG08[7:5]): threshold for releasing the under-voltage state of VDD\_BAT
- BIAS\_OR\_DREFH\_VBAT (REG08[4:2]): under-voltage threshold of VDD\_BAT
- BIAS\_OR\_DREFL\_VDDA (REG09[7:5]): threshold for releasing the under-voltage state of VDD\_ANA
- BIAS\_OR\_DREFH\_VDDA (REG09[4:2]): under-voltage threshold of VDD\_ANA
- BIAS\_OR\_DREFL\_VBAT\_CHARGER (REG10[7:5]): threshold for releasing the under-voltage state of battery connected to VDD\_BAT
- BIAS\_OR\_DREFH\_VBAT\_CHARGER (REG10[4:2]): under-voltage threshold of battery connected to VDD\_BAT

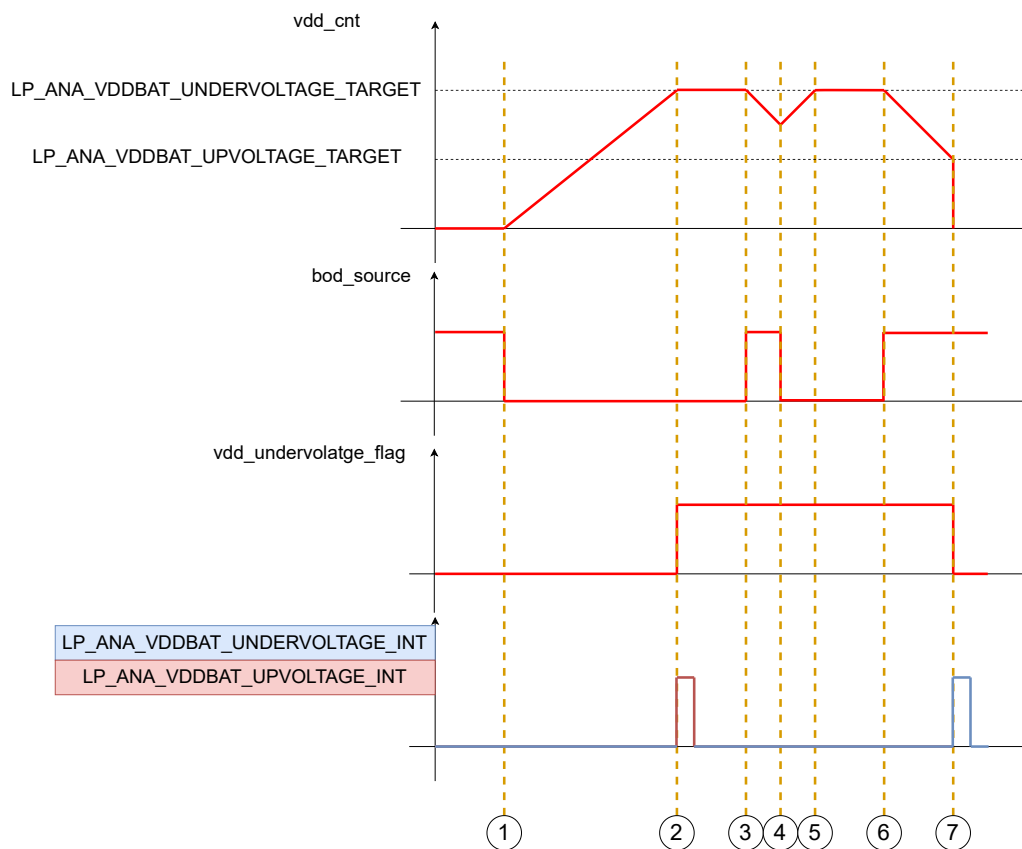
The relationship between the threshold configuration of VBAT/VDDA and voltage is:

- 0: 2.0 V
- 1: 2.1 V
- 2: 2.2 V
- 3: 2.3 V
- 4: 2.4 V
- 5: 2.5 V
- 6: 2.6 V
- 7: 2.7 V

The relationship between the threshold configuration of VBAT\_CHARGER and voltage is:

- 0: 2.2 V

- 1: 2.3 V
  - 2: 2.4 V
  - 3: 2.5 V
  - 4: 2.6 V
  - 5: 2.7 V
  - 6: 2.8 V
  - 7: 2.9 V
- Voltage-monitoring filter  
A dedicated glitch filter is used to process the detection results of VDD\_BAT, as shown in the figure below.



**Figure 23.4-1. Structure of Brown-out Detector**

- `vdd_cnt`: the value of the brown-out counter.
- `bod_source`: the monitored source, which is VDD\_BAT in the figure.
- `vdd_undervoltage_flag`: the under-voltage flag.
- Stage 0 ~ 1: When `bod_source` is 1, it indicates the voltage is normal, and the brown-out counter (`vdd_cnt`) remains 0.
- Stage 1 ~ 2: When `bod_source` is 0, it indicates under-voltage happens, and the brown-out counter will start to count. When the counter value reaches the corresponding threshold, `vdd_undervoltage_flag` will be set to 1, and the interrupt is generated.

- Stage 2 ~ 3: When `vdd_underboltage_flag` is 1, the brown-out counter will stop counting, and stabilize at `LP_ANA_VDDBAT_UNDERVOLTAGE_TARGET`.
- Stage 3 ~ 4 ~ 5: When `bod_source` changes back to 0 for a short period, the value in the brown-out counter will decrease. However, as the value does not reach to `LP_ANA_VDDBAT_UPVOLTAGE_TARGET`, `vdd_underboltage_flag` will not be toggled.
- Stage 6 ~ 7: After `bod_source` remains 0 for a certain period, i.e., the counter value decreases to `LP_ANA_VDDBAT_UPVOLTAGE_TARGET`, `vdd_underboltage_flag` is reset to 0. Then the counter is cleared to 0.

## 23.5 Interrupts

ESP32-P4's brown-out detector can generate the following interrupt and send it to [Interrupt Matrix](#).

- `LP_ANAPERI_INT`

The interrupt is generated by the internal interrupt sources of the brown-out detector.

- `LP_ANA_VDDBAT_CHARGE_UPVOLTAGE_INT`: Triggered when the voltage of `VDD_BAT` increases to the predefined charging threshold.
- `LP_ANA_VDDBAT_CHARGE_UNDERVOLTAGE_INT`: Triggered when the voltage of `VDD_BAT` decreases to the predefined charging threshold.
- `LP_ANA_VDDBAT_UPVOLTAGE_INT`: Triggered when the voltage of `VDD_BAT` increases to the predefined brown-out threshold.
- `LP_ANA_VDDBAT_UNDERVOLTAGE_INT`: Triggered when the voltage of `VDD_BAT` reduces to the predefined brown-out threshold.
- `LP_ANA_BOD_MODE0_INT`: Triggered when the value of the brown-out counter reaches the predefined threshold in Mode 0.

Each interrupt source can be configured by a common set of registers that are described in Chapter [Interrupt Configuration Registers](#). The specific registers can be found in Section [23.7 Register Summary](#).

## 23.6 Programming Procedures

- Mode 0
  - Configure `LP_ANA_BOD_MODE0_RESET_WAIT` to set the counter threshold for triggering the reset signal.
  - Configure `LP_ANA_BOD_MODE0_INTR_WAIT` to set the counter threshold for triggering interrupts.
  - Set `LP_ANA_BOD_MODE0_RESET_SEL` to select the reset way in Mode 0.
  - Configure `LP_ANA_BOD_SOURCE_ENA` to enable the power pin to be detected.
  - Set `LP_ANA_BOD_MODE0_RESET_ENA` to enable the brown-out reset functionality.
  - Set `LP_ANA_BOD_MODE0_INTR_ENA` to enable the noise-filtering counter and interrupts.
- Mode 1

- Configure [LP\\_ANA\\_BOD\\_SOURCE\\_ENA](#) to enable the power pin to be detected.

## 23.7 Register Summary

The addresses in this section are relative to Brown-out Detector base address provided in Table 7.3-2 in Chapter 7 *System and Memory*.

The abbreviations given in Column **Access** are explained in Section *Access Types for Registers*.

| Name                                          | Description                                              | Address | Access |
|-----------------------------------------------|----------------------------------------------------------|---------|--------|
| <a href="#">LP_ANA_BOD_MODE0_CNTL_REG</a>     | Control register of Mode 0                               | 0x0000  | R/W    |
| <a href="#">LP_ANA_VDD_SOURCE_CNTL_REG</a>    | Control register of monitoring sources                   | 0x0008  | varies |
| <a href="#">LP_ANA_VDDBAT_BOD_CNTL_REG</a>    | Control register of brown-out detecting for VDD_BAT      | 0x000C  | varies |
| <a href="#">LP_ANA_VDDBAT_CHARGE_CNTL_REG</a> | Pre-charge control register for under-voltage of VDD_BAT | 0x0010  | varies |
| <a href="#">LP_ANA_INT_RAW_REG</a>            | Raw interrupt status                                     | 0x0020  | RO     |
| <a href="#">LP_ANA_INT_ST_REG</a>             | Interrupt state register                                 | 0x0024  | RO     |
| <a href="#">LP_ANA_INT_ENA_REG</a>            | Interrupt enable register                                | 0x0028  | R/W    |
| <a href="#">LP_ANA_INT_CLR_REG</a>            | Interrupt clear register                                 | 0x002C  | WT     |

## 23.8 Registers

The addresses in this section are relative to Brown-out Detector base address provided in Table 7.3-2 in Chapter 7 *System and Memory*.

For how to program reserved fields, please refer to Section [Programming Reserved Register Field](#).

**Register 23.1. LP\_ANA\_BOD\_MODE0\_CNTL\_REG (0x0000)**

|                            |    |    |    |    |       |  |  |  |  |  |  |  |  |  |  |                             |    |    |  |  |  |  |  |  |  |  |  |  |  |  |   |                            |   |   |   |   |   |   |   |   |   |   |   |       |  |  |   |            |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |                                  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |
|----------------------------|----|----|----|----|-------|--|--|--|--|--|--|--|--|--|--|-----------------------------|----|----|--|--|--|--|--|--|--|--|--|--|--|--|---|----------------------------|---|---|---|---|---|---|---|---|---|---|---|-------|--|--|---|------------|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|----------------------------------|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|
| LP_ANA_BOD_MODE0_RESET_ENA |    |    |    |    |       |  |  |  |  |  |  |  |  |  |  | LP_ANA_BOD_MODE0_RESET_WAIT |    |    |  |  |  |  |  |  |  |  |  |  |  |  |   | LP_ANA_BOD_MODE0_INTR_WAIT |   |   |   |   |   |   |   |   |   |   |   |       |  |  |   | (reserved) |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  | LP_ANA_BOD_MODE0_CLOSE_FLASH_ENA |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |
| LP_ANA_BOD_MODE0_RESET_SEL |    |    |    |    |       |  |  |  |  |  |  |  |  |  |  | LP_ANA_BOD_MODE0_INTR_ENA   |    |    |  |  |  |  |  |  |  |  |  |  |  |  |   | (reserved)                 |   |   |   |   |   |   |   |   |   |   |   |       |  |  |   | (reserved) |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |                                  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |
| LP_ANA_BOD_MODE0_CNTL_CLR  |    |    |    |    |       |  |  |  |  |  |  |  |  |  |  |                             |    |    |  |  |  |  |  |  |  |  |  |  |  |  |   |                            |   |   |   |   |   |   |   |   |   |   |   |       |  |  |   |            |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |                                  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |
| 31                         | 30 | 29 | 28 | 27 |       |  |  |  |  |  |  |  |  |  |  |                             | 18 | 17 |  |  |  |  |  |  |  |  |  |  |  |  | 8 | 7                          | 6 | 5 |   |   |   |   |   |   |   |   |   |       |  |  | 0 |            |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |                                  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |
| 0                          | 0  | 0  | 0  |    | 0x3ff |  |  |  |  |  |  |  |  |  |  |                             |    | 1  |  |  |  |  |  |  |  |  |  |  |  |  |   | 0                          | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | Reset |  |  |   |            |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |                                  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |

**LP\_ANA\_BOD\_MODE0\_CLOSE\_FLASH\_ENA** Configure to close the flash functionality when under-voltage happens in Mode 0.

0: Disable

1: Enable

(R/W)

**LP\_ANA\_BOD\_MODE0\_INTR\_WAIT** Configure the counter threshold that triggers interrupts in Mode 0. (R/W)

**LP\_ANA\_BOD\_MODE0\_RESET\_WAIT** Configure the counter threshold that triggers reset in Mode 0. (R/W)

**LP\_ANA\_BOD\_MODE0\_CNTL\_CLR** Clear counter value in Mode 0. (R/W)

**LP\_ANA\_BOD\_MODE0\_INTR\_ENA** Configure to enable the counter and interrupts in Mode 0.

0: Disable

1: Enable

(R/W)

**LP\_ANA\_BOD\_MODE0\_RESET\_SEL** Configure the reset way when under-voltage happens in Mode 0.

0: Reset the chip

1: Reset the system

(R/W)

**LP\_ANA\_BOD\_MODE0\_RESET\_ENA** Configure to enable reset in Mode 0.

0: Disable

1: Enable

(R/W)



**Register 23.2. LP\_ANA\_VDD\_SOURCE\_CNTL\_REG (0x0008)**

|            |   |   |   |   |                       |    |    |    |   |            |   |   |   |   |                              |    |    |    |    |            |   |   |   |   |                           |   |    |    |   |            |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |     |
|------------|---|---|---|---|-----------------------|----|----|----|---|------------|---|---|---|---|------------------------------|----|----|----|----|------------|---|---|---|---|---------------------------|---|----|----|---|------------|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|-----|
| (reserved) |   |   |   |   | LP_ANA_BOD_SOURCE_ENA |    |    |    |   | (reserved) |   |   |   |   | LP_ANA_VBAT_EVENT_RECORD_CLR |    |    |    |    | (reserved) |   |   |   |   | LP_ANA_VGOOD_EVENT_RECORD |   |    |    |   | (reserved) |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |     |
| 31         |   |   |   |   | 27                    | 26 | 25 | 24 |   |            |   |   |   |   |                              | 19 | 18 | 17 | 16 |            |   |   |   |   |                           |   | 11 | 10 | 9 | 8          |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   | 0 |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |     |
| 0          | 0 | 0 | 0 | 0 | 0                     | 2  | 0  | 0  | 0 | 0          | 0 | 0 | 0 | 0 | 0                            | 0  | 0  | 0  | 0  | 0          | 0 | 0 | 0 | 0 | 0                         | 0 | 0  | 0  | 0 | 0          | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0</ |

**LP\_ANA\_VGOOD\_EVENT\_RECORD** Flags for indicating pins whose voltage is lower than the threshold.

0: Under-voltage does not happen on either VDD\_BAT or VDD\_ANA;

1: Under-voltage happens on VDD\_BAT;

2: Under-voltage happens on VDD\_ANA;

3: Under-voltage happens on both VDD\_BAT and VDD\_ANA.

(RO)

**LP\_ANA\_VBAT\_EVENT\_RECORD\_CLR** Clear the flags.

0: Not clear

1: Clear the brown-out flag of VDD\_BAT

2: Clear the brown-out flag of VDD\_ANA

3: No effect

(WT)

**LP\_ANA\_BOD\_SOURCE\_ENA** Configure to detect monitoring sources.

0: Not detecting any pins;

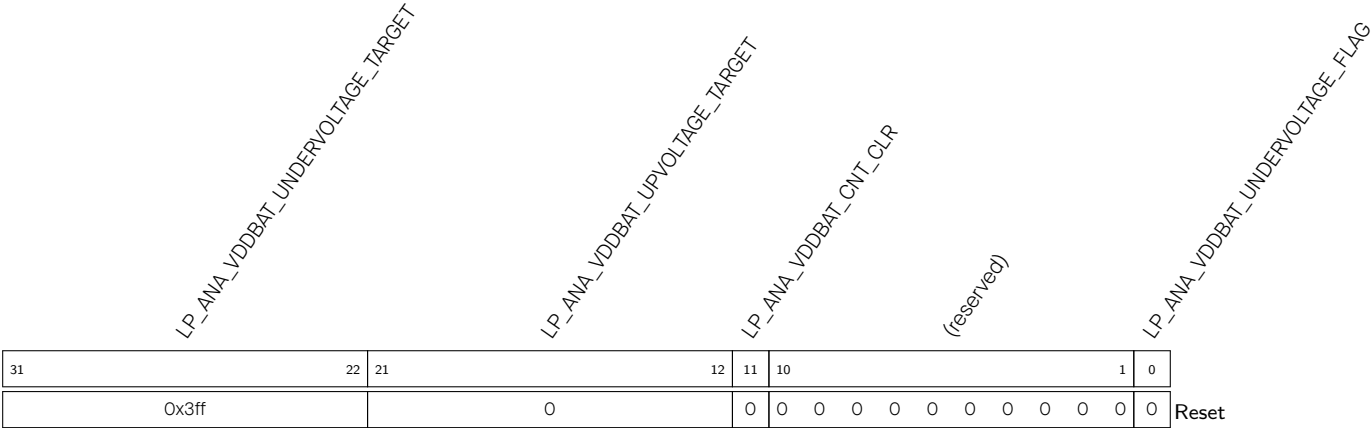
1: Only detect VDD\_BAT;

2: Only detect VDD\_ANA;

3: Detect both VDD\_BAT and VDD\_ANA.

(R/W)

Register 23.3. LP\_ANA\_VDDBAT\_BOD\_CNTL\_REG (0x000C)



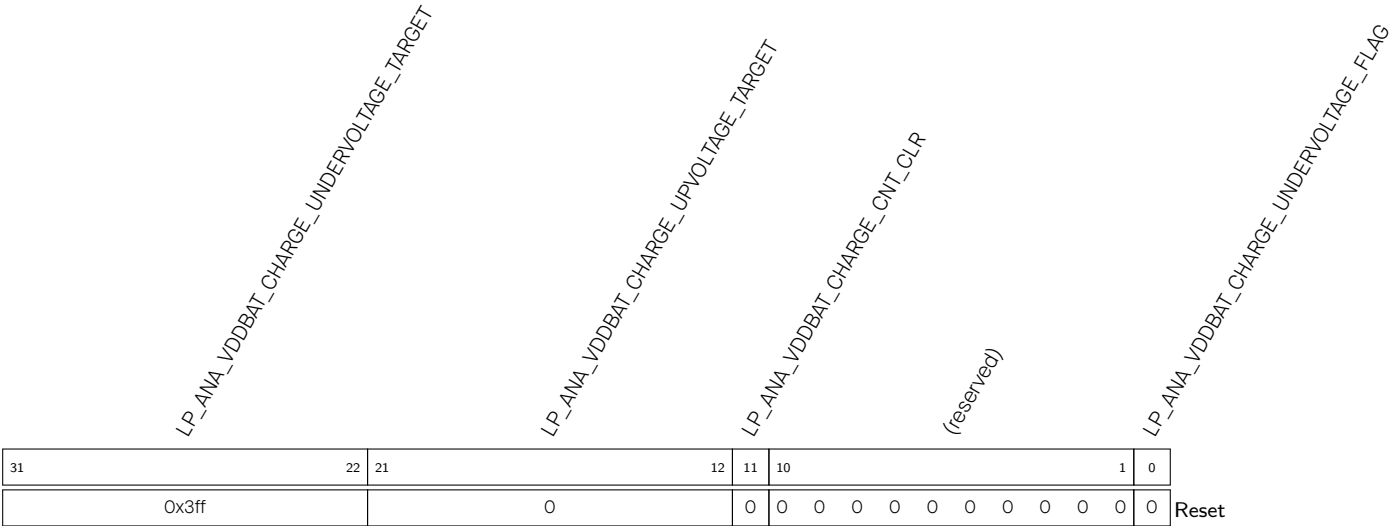
**LP\_ANA\_VDDBAT\_UNDERVOLTAGE\_FLAG** Flag indicating that under-voltage occurs on VDD\_BAT. (RO)

**LP\_ANA\_VDDBAT\_CNTL\_CLR** Clear the brown-out counter of VDD\_BAT. (WT)

**LP\_ANA\_VDDBAT\_UPVOLTAGE\_TARGET** Configure the voltage-recovery threshold for the brown-out counter of VDD\_BAT. When the counter decreases to this threshold, [LP\\_ANA\\_VDDBAT\\_UNDERVOLTAGE\\_FLAG](#) is cleared to 0. (R/W)

**LP\_ANA\_VDDBAT\_UNDERVOLTAGE\_TARGET** Configure the brown-out threshold for the brown-out counter of VDD\_BAT. When the counter increases to this threshold, [LP\\_ANA\\_VDDBAT\\_UNDERVOLTAGE\\_FLAG](#) is set to 1. (R/W)

Register 23.4. LP\_ANA\_VDDBAT\_CHARGE\_CNTL\_REG (0x0010)



**LP\_ANA\_VDDBAT\_CHARGE\_UNDERVOLTAGE\_FLAG** Flag indicating that VDD\_BAT is charging. (RO)

**LP\_ANA\_VDDBAT\_CHARGE\_CNT\_CLR** Clear the brown-out counter of VDD\_BAT. (WT)

**LP\_ANA\_VDDBAT\_CHARGE\_UPVOLTAGE\_TARGET** Configure the voltage-recovery threshold for the brown-out counter of VDD\_BAT. When the counter decreases to this threshold, [LP\\_ANA\\_VDDBAT\\_CHARGE\\_UNDERVOLTAGE\\_FLAG](#) is cleared to 0. (R/W)

**LP\_ANA\_VDDBAT\_CHARGE\_UNDERVOLTAGE\_TARGET** Configure the brown-out threshold for the brown-out counter of VDD\_BAT. When the counter increases to this threshold, [LP\\_ANA\\_VDDBAT\\_CHARGE\\_UNDERVOLTAGE\\_FLAG](#) is set to 1. (R/W)

Register 23.5. LP\_ANA\_INT\_RAW\_REG (0x0020)

|                                                                                                                                                                                                            |    |    |    |    |    |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |       |  |
|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----|----|----|----|----|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|-------|--|
| LP_ANA_BOD_MODE0_INT_RAW<br>LP_ANA_VDDBAT_UNDERVOLTAGE_INT_RAW<br>LP_ANA_VDDBAT_UPVOLTAGE_INT_RAW<br>LP_ANA_VDDBAT_CHARGE_UNDERVOLTAGE_INT_RAW<br>LP_ANA_VDDBAT_CHARGE_UPVOLTAGE_INT_RAW<br><br>(reserved) |    |    |    |    |    |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |       |  |
| 31                                                                                                                                                                                                         | 30 | 29 | 28 | 27 | 26 |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   | 0     |  |
| 0                                                                                                                                                                                                          | 0  | 0  | 0  | 0  | 0  | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | Reset |  |

- LP\_ANA\_VDDBAT\_CHARGE\_UPVOLTAGE\_INT\_RAW The interrupt is triggered when the voltage of VDD\_BAT increases to the charging threshold. (RO)
- LP\_ANA\_VDDBAT\_CHARGE\_UNDERVOLTAGE\_INT\_RAW The interrupt is triggered when the voltage of VDD\_BAT decreases to the charging threshold. (RO)
- LP\_ANA\_VDDBAT\_UPVOLTAGE\_INT\_RAW The interrupt is triggered when the voltage of VDD\_BAT increases to the brown-out threshold. (RO)
- LP\_ANA\_VDDBAT\_UNDERVOLTAGE\_INT\_RAW The interrupt is triggered when the voltage of VDD\_BAT decreases to the brown-out threshold. (RO)
- LP\_ANA\_BOD\_MODE0\_INT\_RAW Brown-out interrupt in Mode 0. (RO)

### Register 23.6. LP\_ANA\_INT\_ST\_REG (0x0024)

[illegible]

**LP\_ANA\_VDDBAT\_CHARGE\_UPVOLTAGE\_INT\_ST** The interrupt is triggered and sent to CPU when the voltage of VDD\_BAT increases to the charging threshold. (RO)

**LP\_ANA\_VDDBAT\_CHARGE\_UNDERVOLTAGE\_INT\_ST** The interrupt is triggered and sent to CPU when the voltage of VDD\_BAT decreases to the charging threshold. (RO)

**LP\_ANA\_VDDBAT\_UPVOLTAGE\_INT\_ST** The interrupt is triggered and sent to CPU when the voltage of VDD\_BAT increases to the brown-out threshold. (RO)

**LP\_ANA\_VDDBAT\_UNDERVOLTAGE\_INT\_ST** The interrupt is triggered and sent to CPU when the voltage of VDD\_BAT decreases to the brown-out threshold. (RO)

**LP\_ANA\_BOD\_MODE0\_INT\_ST** The brown-out interrupt in Mode 0 is triggered and sent to CPU.  
(RO)

Register 23.7. LP\_ANA\_INT\_ENA\_REG (0x0028)

|                                                                                                                                                                                                            |    |    |    |    |    |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |       |
|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----|----|----|----|----|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|-------|
| LP_ANA_BOD_MODE0_INT_ENA<br>LP_ANA_VDDBAT_UNDERVOLTAGE_INT_ENA<br>LP_ANA_VDDBAT_UPVOLTAGE_INT_ENA<br>LP_ANA_VDDBAT_CHARGE_UNDERVOLTAGE_INT_ENA<br>LP_ANA_VDDBAT_CHARGE_UPVOLTAGE_INT_ENA<br><br>(reserved) |    |    |    |    |    |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |       |
| 31                                                                                                                                                                                                         | 30 | 29 | 28 | 27 | 26 |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   | 0 |   |       |
| 0                                                                                                                                                                                                          | 0  | 0  | 0  | 0  | 0  | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | Reset |

**LP\_ANA\_VDDBAT\_CHARGE\_UPVOLTAGE\_INT\_ENA** Configure whether to enable the interrupt which is triggered when the voltage of VDD\_BAT increases to the charging threshold.

0: No

1: Yes

(R/W)

**LP\_ANA\_VDDBAT\_CHARGE\_UNDERVOLTAGE\_INT\_ENA** Configures whether to enable the interrupt which is triggered when the voltage of VDD\_BAT reduces to the charging threshold.

0: Yes

1: No

(R/W)

**LP\_ANA\_VDDBAT\_UPVOLTAGE\_INT\_ENA** Configures whether to enable the interrupt which is triggered when the voltage of VDD\_BAT increases to the brown-out threshold.

0: Yes

1: No

(R/W)

**LP\_ANA\_VDDBAT\_UNDERVOLTAGE\_INT\_ENA** Configures whether to enable the interrupt which is triggered when the voltage of VDD\_BAT reduces to the brown-out threshold.

0: No

1: Yes

(R/W)

**LP\_ANA\_BOD\_MODE0\_INT\_ENA** Configures whether to enable the brown-out interrupt in Mode 0.

0: No

1: Yes

(R/W)

### Register 23.8. LP\_ANA\_INT\_CLR\_REG (0x002C)

Reset register (0x00000000) bit fields:

- Bit 31: LP\_ANA\_BOD\_MODE0\_INT\_CLR
- Bit 30: LP\_ANA\_VDDBAT\_UNDERVOLTAGE\_INT\_CLR
- Bit 29: LP\_ANA\_VDDBAT\_CHARGE\_UPVOLTAGE\_INT\_CLR
- Bit 28: LP\_ANA\_VDDBAT\_CHARGE\_UPVOLTAGE\_INT\_CLR
- Bit 27: LP\_ANA\_VDDBAT\_CHARGE\_UPVOLTAGE\_INT\_CLR
- Bit 26: (reserved)
- Bits 0-25: 0

**LP\_ANA\_VDDBAT\_CHARGE\_UPVOLTAGE\_INT\_CLR** Configures whether to clear the interrupt triggered by the VDD\_BAT voltage increasing to the charging threshold. (WT)

**LP\_ANA\_VDDBAT\_CHARGE\_UNDERVOLTAGE\_INT\_CLR** Configures whether to clear the interrupt triggered by the VDD\_BAT voltage reducing to the charging threshold. (WT)

**LP\_ANA\_VDDBAT\_UPVOLTAGE\_INT\_CLR** Configures whether to clear the interrupt triggered by the VDD\_BAT voltage increasing to the brown-out threshold. (WT)

**LP\_ANA\_VDDBAT\_UNDERVOLTAGE\_INT\_CLR** Configures whether to clear the interrupt triggered by the VDD\_BAT voltage reducing to the brown-out threshold. (WT)

LP\_ANA\_BOD\_MODE0\_INT\_ST Configures whether to clear the brown-out interrupt in Mode 0.  
(WT)